

AI FOUNDATIONS

1. Artificial Intelligence (AI)

Artificial Intelligence (AI) is a branch of computer science that focuses on creating systems that can perform tasks that normally require human intelligence. These tasks include learning, reasoning, problem-solving, understanding language, recognizing images, making decisions, and generating content. For example, a voice assistant understanding your question, a recommendation system suggesting movies, and an AI detecting objects in an image are all applications of AI.

2. Why Do We Need AI?

AI is used because computers can process huge amounts of data much faster than humans and can automate tasks that would otherwise require a lot of human effort. AI can help organizations make predictions, find patterns, automate repetitive work, improve decision-making, and provide personalized services. For example, banks use AI to detect suspicious transactions, while shopping platforms use AI to recommend products.

3. Human Intelligence vs Artificial Intelligence

Human intelligence comes naturally from the human brain and involves abilities such as emotions, common sense, creativity, experience, and reasoning. **Artificial intelligence** tries to reproduce some of these abilities using computers and mathematical models. AI can be extremely fast at processing data, but it does not possess human intelligence in the complete sense. Current AI systems are

generally designed for specific tasks rather than having human-like understanding of everything.

4. Types of AI Based on Capability

AI is commonly divided into **Narrow AI**, **General AI**, and **Super AI** based on its theoretical level of intelligence.

Narrow AI (ANI) is AI designed to perform a specific task or a limited range of tasks. Examples include recommendation systems, spam filters, facial recognition, and language models. Most AI systems we use today fall into this category.

Artificial General Intelligence (AGI) refers to a hypothetical AI that would have the ability to understand, learn, and perform a wide variety of intellectual tasks at a human-like level. AGI has not been achieved.

Artificial Super Intelligence (ASI) refers to a hypothetical form of AI that would surpass human intelligence across essentially all intellectual areas. It remains a theoretical concept.

5. Types of AI Based on Functionality

AI can also be classified according to how it behaves. **Reactive machines** respond to current inputs without maintaining a meaningful memory of previous experiences. **Limited-memory AI** can use information from previous observations for a limited period and is common in modern AI applications. More advanced theoretical categories include **Theory of Mind AI**, which would understand human emotions and intentions, and **Self-aware AI**, which would possess self-awareness. The last two are not currently available.

MACHINE LEARNING

6. What is Machine Learning?

Machine Learning (ML) is a subset of AI in which computers learn patterns from data and use those patterns to make predictions or decisions instead of being explicitly programmed with every rule. For example, instead of manually writing rules for identifying spam emails, we can provide many examples of spam and normal emails and allow an ML model to learn the patterns.

7. AI vs Machine Learning

AI is the **broader concept** of making machines perform tasks associated with intelligence, while Machine Learning is one of the major approaches used to achieve AI. In simple terms, **AI is the larger field and ML is a part of AI**. For example, an AI-based spam detector can use machine learning to learn from previous emails.

Remember:

AI → broader field

ML → subset of AI

8. Traditional Programming vs Machine Learning

In traditional programming, we provide **rules and data**, and the computer produces an output.

Traditional programming:

Rules + Data → Output

In machine learning, we provide **data and expected outputs**, and the system learns the rules or patterns.

Machine Learning:

Data + Expected Output → Model

The trained model can then be used with new data to produce predictions.

9. What is Data?

Data is the information used by an AI or ML system to learn patterns or make decisions. Data can be text, numbers, images, audio, video, sensor readings, documents, or user activity. For example, a house-price prediction system may use data such as area, number of rooms, location, and previous selling prices.

10. Dataset

A **dataset** is a collection of data used for analysis or machine learning. It usually contains multiple examples or records. For example, a student-performance dataset could contain information about study hours, attendance, previous marks, and final results for thousands of students. The quality and quantity of the dataset strongly affect the performance of an ML model.

11. Features

Features are the individual pieces of information used by a machine-learning model to make a prediction. For example, if we want to predict whether a student will pass, features could include study hours, attendance, assignment completion, and previous marks. Features are essentially the **inputs given to the model**.

12. Label

A **label** is the correct or expected output associated with an example in a dataset. For example, if we are predicting whether a student passes an exam, the label could be **Pass** or **Fail**. Labels are mainly used in supervised learning, where the model learns from examples that already have known answers.

13. Model

A **model** is a trained computational system that has learned patterns from data and can use those patterns to make predictions or generate outputs. For example, after being trained on many examples of houses and their prices, a model can predict the approximate price of a new house.

14. Training

Training is the process through which a machine-learning model learns patterns from data. During training, the model makes predictions, compares them with the expected results, calculates its error, and adjusts its internal parameters to improve its predictions. This process is repeated many times.

15. Inference

Inference is the process of using a trained model to make a prediction or produce an output for new data. Training is when the model learns, while inference is when we use what the model has learned. For example, training a model on thousands of cat and dog images is training; giving it a new image and asking it to identify the animal is inference.

Remember:

Training = Learning

Inference = Using the learned model

TYPES OF MACHINE LEARNING

16. Supervised Learning

Supervised learning is a type of machine learning where the model is trained using data that already has correct answers or labels. The model learns the relationship between inputs and outputs and then uses that knowledge to predict outputs for new inputs. Examples include predicting house prices and classifying emails as spam or not spam.

17. Classification

Classification is a supervised learning problem where the output belongs to a category or class. For example, an email can be classified as **Spam** or **Not Spam**, and a medical image can be classified into different categories. The output is not normally a continuous numerical value; it represents a class.

18. Regression

Regression is a supervised learning problem where the model predicts a numerical value. For example, predicting the price of a house, temperature, salary, or sales amount are regression problems. The output can take different numerical values.

Easy difference:

Classification → category

Regression → numerical value

19. Unsupervised Learning

Unsupervised learning is a type of machine learning where the data does not contain predefined labels. The model tries to discover hidden patterns, structures, or groups within the data on its own. For example, a shopping company can use unsupervised learning to group customers based on their purchasing behavior.

20. Clustering

Clustering is an unsupervised learning technique that groups similar data points together. For example, a company could use customer information to create groups such as customers who frequently purchase electronics, customers who mostly purchase clothing, and customers who purchase both. The groups are discovered from the data rather than being provided beforehand.

21. Reinforcement Learning

Reinforcement Learning (RL) is a type of machine learning in which an **agent** learns by interacting with an environment. The agent performs actions and receives rewards or penalties depending on the result. Over time, it learns which actions are more useful.

Reinforcement learning is commonly associated with areas such as game-playing, robotics, and decision-making.

22. Agent

An **agent** is the entity that makes decisions or takes actions in an environment. In reinforcement learning, the agent observes the current situation, chooses an action, and receives a reward or penalty. For example, in a game, the AI player can act as an agent.

23. Environment

The **environment** is the world or system in which an AI agent operates. It provides information to the agent and responds to the agent's actions. In a game, the game itself is the environment; in robotics, the physical surroundings can be the environment.

IMPORTANT ML CONCEPTS

24. Training Data

Training data is the portion of a dataset used to teach the model. The model analyzes this data and adjusts its internal parameters to learn patterns. A model generally needs enough representative training data to learn effectively.

25. Testing Data

Testing data is data that the model has not used during training and is used to evaluate how well the trained model performs on unseen examples. Testing helps us understand whether the model can generalize beyond the examples it was trained on.

26. Validation Data

Validation data is used during model development to help select or tune the model and its settings. It is different from the final test data because the test set is ideally kept untouched until the final evaluation.

A simple division is:

Training → Learn

Validation → Tune/Choose

Testing → Final Evaluation

27. Overfitting

Overfitting happens when a model learns the training data too closely, including its noise and unimportant details. As a result, it performs very well on training data but poorly on new, unseen data. It is similar to a student memorizing answers instead of understanding the concept.

28. Underfitting

Underfitting happens when a model is too simple to properly learn the important patterns in the training data. It performs poorly on both training data and new data. It is similar to a student who has not studied enough to understand the topic.

29. Generalization

Generalization refers to a model's ability to perform well on new, unseen data. A good ML model should not simply memorize the

training examples; it should learn useful patterns that also work on new examples.

30. Parameters

Parameters are internal values of a model that are learned during training. They control how the model transforms its inputs into predictions. In neural networks, weights and biases are examples of parameters. Large modern AI models can contain millions or billions of learned parameters.

31. Hyperparameters

Hyperparameters are settings chosen before or during the training process rather than learned directly from the training data in the same way as model parameters. Examples include learning rate, batch size, and number of training epochs. They influence how the model learns.

Easy difference:

Parameters → learned by the model

Hyperparameters → set/configured by us

32. Algorithm

An **algorithm** is a step-by-step method used to solve a problem. In machine learning, algorithms provide the mathematical procedures used to learn patterns from data. Examples include decision trees, linear regression, logistic regression, k-means, and neural-network training methods.

DEEP LEARNING

33. What is Deep Learning?

Deep Learning is a subset of machine learning that uses artificial neural networks with multiple layers to learn complex patterns from large amounts of data. It is especially useful for tasks involving images, speech, text, and other complex data. Modern generative AI and many advanced computer-vision systems rely heavily on deep learning.

34. Machine Learning vs Deep Learning

Machine learning includes many different algorithms and approaches, while deep learning is a specific area of machine learning based primarily on multi-layer neural networks. Traditional ML often requires humans to design or select useful features, while deep learning can often learn useful representations automatically from raw data when enough data and computing resources are available.

NEURAL NETWORKS

35. Neural Network

A **neural network** is a computational model made up of interconnected units called neurons or nodes. It generally contains an input layer, one or more hidden layers, and an output layer. The network learns by adjusting the weights of connections between neurons during training.

36. Neuron

An artificial **neuron** receives input values, combines them using weights and a bias, and passes the result through an activation function to produce an output. Many such neurons are connected together to form a neural network.

37. Input Layer

The **input layer** receives the features or input information given to a neural network. For example, if a model receives information about study hours, attendance, and previous marks, these values enter through the input layer.

38. Hidden Layer

A **hidden layer** is an intermediate layer between the input and output layers. It performs transformations on the data and helps the network learn complex patterns. Deep learning models usually contain many such layers.

39. Output Layer

The **output layer** produces the final result of the neural network. For example, it might output whether an email is spam, the predicted price of a house, or the probability of different classes.

40. Weight

A **weight** represents the importance or strength of a connection between neurons. During training, the model adjusts weights so that its predictions become more accurate. A larger learned weight can indicate that a particular input has a stronger influence on a particular calculation, although its exact interpretation depends on the model.

41. Bias

A **bias** is another learned value used by a neuron to shift its output. It gives the model additional flexibility when learning patterns. In a simple mathematical form, a neuron can calculate something like:

$$\text{Output} = (\text{Input} \times \text{Weight}) + \text{Bias}$$

An activation function may then be applied to this result.

42. Activation Function

An **activation function** determines how a neuron's calculated value is transformed before being passed forward. It introduces non-linearity, which allows neural networks to learn complex relationships. Common activation functions include **ReLU, Sigmoid, and Tanh**.

43. Loss Function

A **loss function** measures how different the model's prediction is from the correct answer. A smaller loss generally means the prediction is closer to the desired result. During training, the model tries to adjust its parameters to reduce the loss.

44. Optimization

Optimization is the process of adjusting a model's parameters to reduce the loss and improve its performance. Optimization algorithms determine how the model should change its parameters during training. **Gradient Descent** is one of the most important optimization concepts in machine learning and deep learning.

45. Backpropagation

Backpropagation is a method used to calculate how the model's parameters contributed to the error and determine how they should be adjusted. The error information is propagated backward through the neural network, allowing the model to update its weights and biases during training.

GENERATIVE AI

46. Generative AI

Generative AI is a type of AI that can generate new content based on patterns learned from data. Unlike systems that only classify or predict, generative models can produce content such as text, images, audio, video, and code. Examples include AI systems that write an email, generate an image from a description, or create code from a natural-language request.

47. Generative AI vs Traditional AI

Traditional AI is often designed to **analyze, classify, predict, or make decisions**, while Generative AI focuses on **creating new content**. For example, a traditional AI system might determine whether an image contains a cat, while a generative AI system could create a new cat image based on a text description.

48. Large Language Model (LLM)

A **Large Language Model (LLM)** is a machine-learning model designed to process and generate human language. LLMs learn statistical patterns from large collections of text and can perform tasks such as answering questions, summarizing text, translating languages, writing content, and generating code. Modern LLMs are generally based on deep-learning architectures, particularly Transformer-based architectures.

49. Natural Language Processing (NLP)

Natural Language Processing (NLP) is a field of AI concerned with enabling computers to work with human language. NLP includes tasks such as text classification, translation, sentiment analysis, summarization, question answering, speech-related language processing, and information extraction. LLMs are a major modern development within NLP.

50. Token

A **token** is a unit of text processed by a language model. A token can represent a whole word, part of a word, punctuation, or another piece of text depending on the model's tokenizer. Before an LLM processes text, the text is generally converted into tokens, which are then represented numerically for the model.

51. Tokenization

Tokenization is the process of breaking text into tokens that a language model can process. For example, a sentence may be divided into words or smaller pieces. The exact way text is tokenized depends on the tokenizer and model.

52. Embeddings

Embeddings are numerical representations of data such as words, sentences, documents, or images. They represent information as vectors in a mathematical space so that items with similar meanings or characteristics can be represented as being closer to one another. Embeddings are widely used in search, recommendation systems, and Retrieval-Augmented Generation.

53. Vector

A **vector** is a list of numerical values representing some information in a mathematical space. In AI, vectors are often used to represent features, words, sentences, images, or other objects. For example, an embedding can be represented as a high-dimensional vector.

54. Transformer

A **Transformer** is a deep-learning architecture that became extremely important in modern NLP and Generative AI. It uses a mechanism called **attention** to determine which parts of the input are important in relation to one another. Transformers are the foundation of many modern language models because they can efficiently process relationships between different parts of text.

55. Attention

Attention is a mechanism that allows a model to focus on different parts of the input when processing a particular piece of information. For example, when understanding a sentence, the model can determine which other words are most relevant to the current word. This helps language models capture relationships and context within text.

56. Prompt

A **prompt** is the instruction or input given to an AI system to tell it what you want it to do. For example:

"Explain machine learning in simple language with an example."

is a prompt. A clear prompt generally gives the model better information about the desired task, context, format, or constraints.

57. Prompt Engineering

Prompt engineering is the practice of designing and improving prompts so that an AI model produces more useful and reliable results. It can involve clearly defining the task, providing context, specifying the desired format, giving examples, and setting constraints.

58. Context Window

A **context window** is the amount of information a model can consider as context while generating or processing a response. It can include the user's prompt, previous conversation, documents, and other provided information. The size of the context window depends on the particular model.

59. Hallucination

An **AI hallucination** occurs when an AI system produces information that sounds convincing but is incorrect, unsupported, or invented. This is an important limitation of generative AI. Therefore, important information should be verified rather than blindly trusting an AI-generated answer.

60. Fine-Tuning

Fine-tuning is the process of further training an already trained model on a more specific dataset so that it becomes better suited to a particular task, domain, or style. For example, a general language model could be fine-tuned for a specialized task using carefully prepared examples.

61. RAG

RAG stands for Retrieval-Augmented Generation. It combines information retrieval with generative AI. Instead of relying only on what the model learned during its original training, a RAG system retrieves relevant information from an external knowledge source and provides that information to the model as context before generating an answer.

A simplified process is:

User Question → Retrieve Relevant Information → Give Context to Model → Generate Answer

RAG is commonly used for question-answering systems over company documents, knowledge bases, and other private or frequently updated information.

62. AI Agent

An **AI agent** is a system designed to pursue a goal by observing information, reasoning about what to do, and taking actions, often with access to tools or external systems. A simple chatbot mainly responds to prompts, while an agentic system may be able to plan steps, call tools, retrieve information, and perform actions depending on its design and permissions.

COMPUTER VISION

63. Computer Vision

Computer Vision is a field of AI that enables computers to process and understand information from images and videos. Common applications include face detection, object detection, medical-image analysis, autonomous vehicles, quality inspection, and image classification.

64. Image Classification

Image classification means assigning an image to one or more predefined categories. For example, an AI system might classify an image as **cat**, **dog**, or **bird**.

65. Object Detection

Object detection identifies objects in an image and usually determines their locations using bounding boxes. For example, a road image might contain several cars, pedestrians, and traffic signs, and an object-detection model can identify and locate each of them.

RESPONSIBLE AI

66. Responsible AI

Responsible AI means designing and using AI systems in ways that are safe, fair, transparent, reliable, privacy-conscious, and beneficial to people. AI systems should be evaluated not only for performance but also for possible risks such as bias, privacy problems, misinformation, and misuse.

67. AI Bias

AI bias occurs when an AI system produces systematically unfair or unbalanced results, often because of biased or unrepresentative training data, model design, or the way the system is used. For example, if training data does not adequately represent certain groups, the resulting model may perform worse for those groups.

68. AI Privacy

AI systems often process large amounts of information, some of which may be sensitive. **AI privacy** involves protecting personal information, controlling who can access data, using data appropriately, and avoiding unnecessary collection or exposure of personal information.

69. Explainable AI

Explainable AI (XAI) focuses on making AI decisions or predictions easier for humans to understand. This is particularly important in areas such as healthcare, finance, and other situations where people may need to understand why an AI system produced a particular result.

70. AI Limitations

AI systems have several limitations. They can make incorrect predictions, depend heavily on the quality of their data, reproduce biases, misunderstand context, generate hallucinations, and sometimes require significant computational resources. AI should therefore be treated as a powerful tool rather than an infallible source of truth.

HOW AN AI PROJECT GENERALLY WORKS

71. AI Project Lifecycle

A typical AI/ML project starts by defining the problem, collecting relevant data, cleaning and preparing that data, selecting an appropriate approach, training a model, evaluating its performance, improving it, and finally deploying it for real-world use. After deployment, the model should continue to be monitored because data and real-world conditions can change.

A simple flow is:

Problem → Data → Preparation → Training → Evaluation → Deployment → Monitoring

72. Model Evaluation

Model evaluation is the process of measuring how well a model performs on a specific task. Different tasks require different evaluation metrics. For example, classification can use metrics such as accuracy, precision, recall, and F1-score, while regression can use metrics such as MAE and MSE.

73. Accuracy

Accuracy measures the proportion of predictions that are correct out of all predictions.

Formula:

$$\text{Accuracy} = \text{Correct Predictions} / \text{Total Predictions}$$

For example, if a model makes 90 correct predictions out of 100, its accuracy is 90%.

However, accuracy can be misleading when classes are highly imbalanced, so other metrics may be more appropriate.

74. Precision

Precision tells us, among the cases the model predicted as positive, how many were actually positive. It is especially useful when false positives are costly.

For example, if an email system marks 100 emails as spam but only 80 are actually spam, its precision is 80%.

75. Recall

Recall tells us, among all the actual positive cases, how many the model successfully identified. It is especially important when missing a positive case is costly.

For example, in a disease-detection system, recall can be important because we may want to identify as many actual positive cases as possible.

76. F1-Score

F1-score combines precision and recall into a single metric using their harmonic mean. It is useful when we want a balance between precision and recall, especially when the dataset is imbalanced.